Towards Text-Line Segmentation of Historical Documents Using Graph Neural Networks

Kartik Chincholikar Kaushik Gopalan Mihir Hasabnis

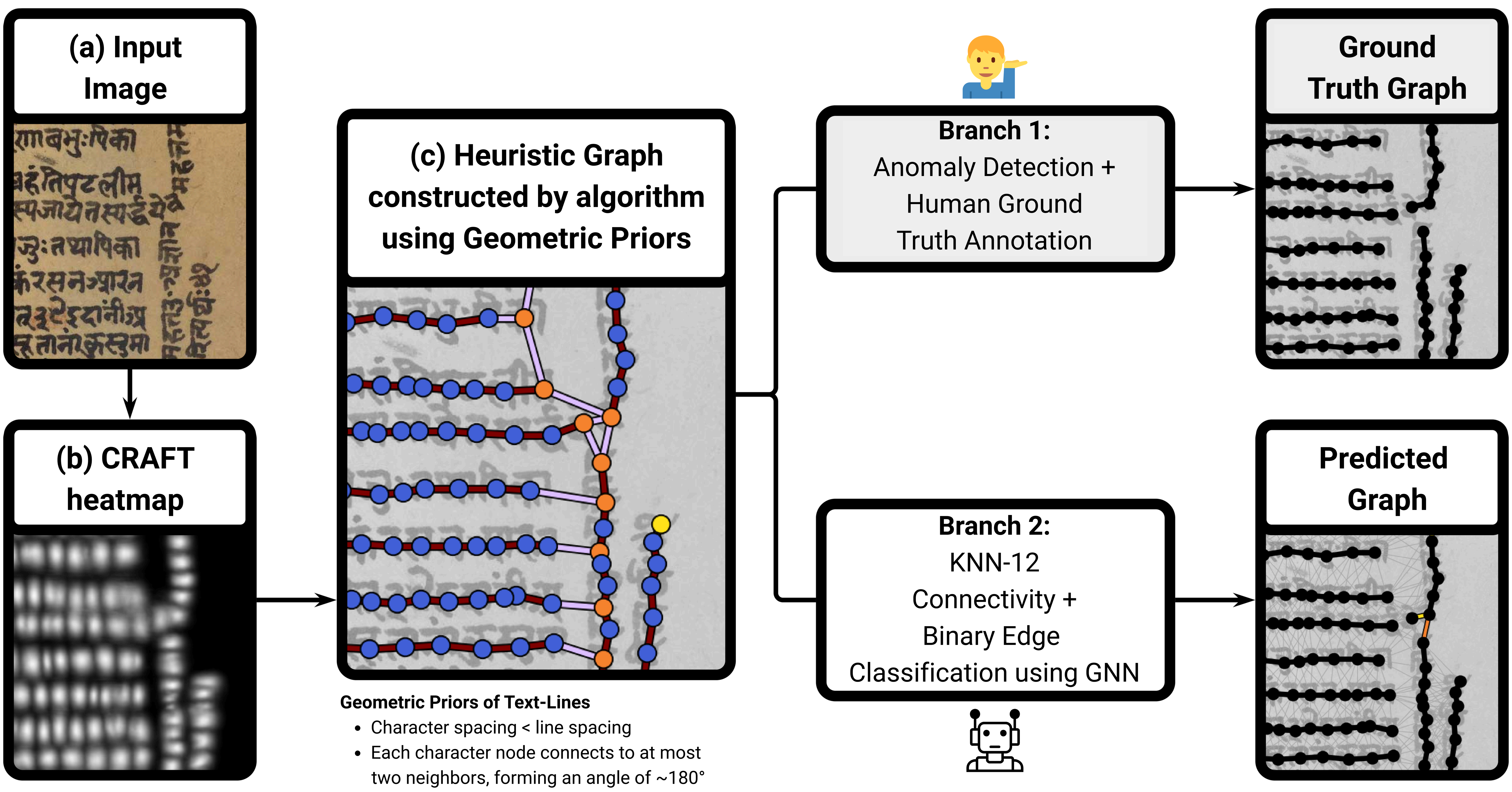
FLAME University, Pune, India

Introduction. We formulate the task of text-line segmentation from historical documents in a GNN-friendly way, in which characters are represented as nodes and edges connect each character to its previous and next characters on the text line. This converts the **[text-line segmentation task]** into a **[character segmentation + binary edge classification task]**, enabling training on large-scale synthetic data simulating complex layouts.

kartik.chincholikar@flame.edu.in

KartikC14

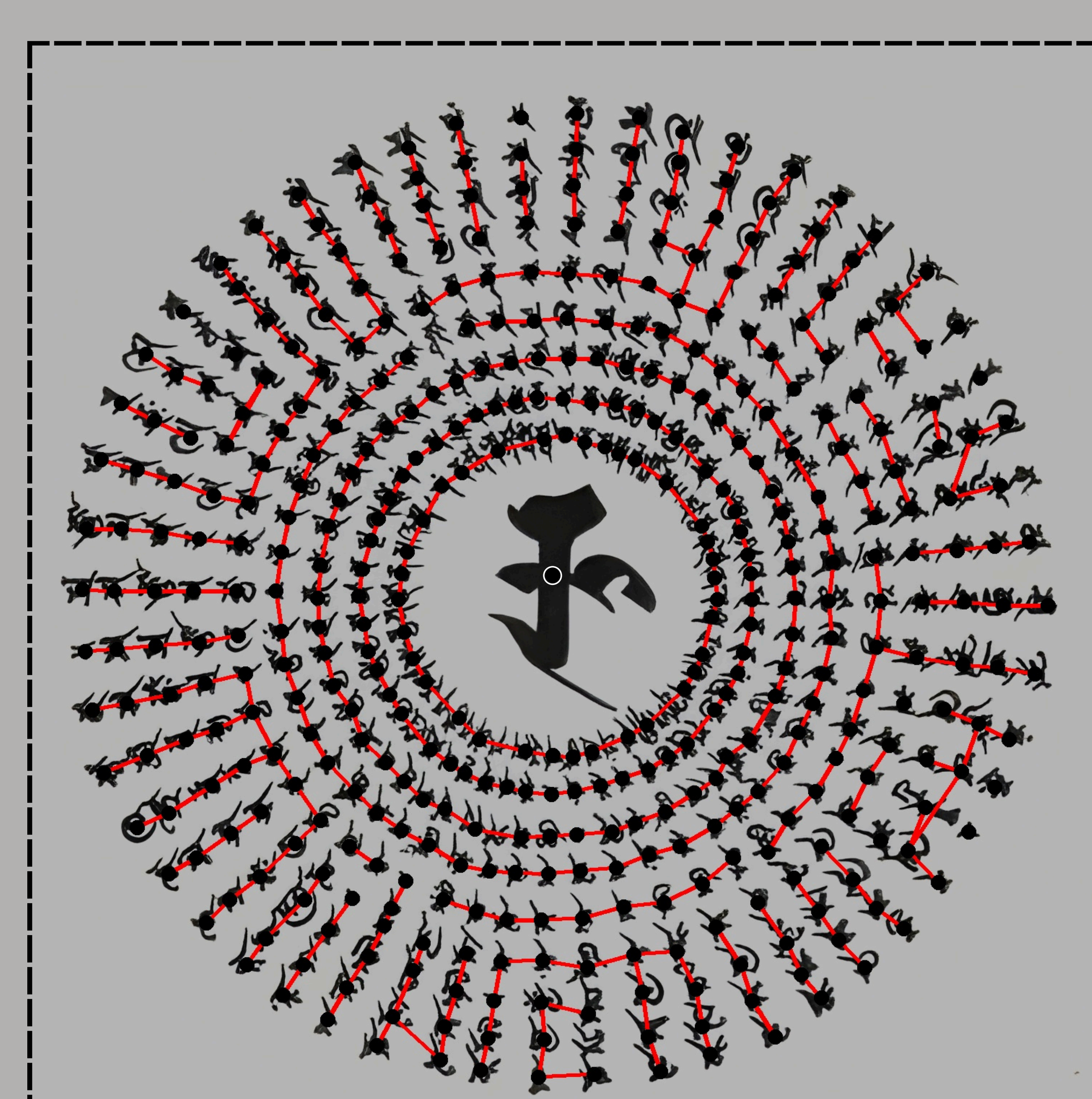
Method. An original manuscript image **(a)** is converted into a heatmap using CRAFT **(b)**. From this, an algorithm that incorporates geometric priors of text-lines creates a heuristic graph, in which the colors of nodes and edges are obtained from the algorithm **(c)**. These colors are informative and crucial for **GCN and GAT** to perform the **binary edge classification** task successfully, but are less important for **MPNN and SplineCNN**, suggesting that the latter may be *implicitly and neurally* learning the geometric priors of text-lines from the raw continuous geometric graph features (see ablation study). **Branch 1** was used for crude semi-automatic data annotation 💁‍♂️, while **Branch 2** automatically performs the task using GNNs 🤖.



Qualitative Out-of-Distribution Evaluation. We visualize the predictions of out-of-the-box competing methods **SeamFormer**, **Doc-UFCN**, and the **Proposed-method** on a Siddham Manuscript page having a *radial layout (to be read clockwise)*. As training data of the Proposed-method (including the synthetic data) DID NOT contain such radial layouts, this figure highlights the generalizability of the Proposed-method. Furthermore, the prediction of the Proposed-method seems to be ***more salvageable***, allowing the user to manually correct mistakes by adding/deleting edges and nodes as required.

SeamFormer. End to End Dense Pixel-Level text-line segmentation

Doc-UFCN. End to End Dense Pixel-Level text-line segmentation



Proposed Method. Dense Pixel-Level character segmentation + GNN Binary Edge Classification

Competitive Accuracy, better Consistency* ✅ *on selected benchmark datasets, and the new dataset introduced

Beats end-to-end Gemini on Downstream OCR* ✅ *for Sanskrit Manuscripts, the traditional OCR pipeline, outperforms MultiModel LLMs

Trainable on Synthetic Layout Data ✅

Dependent on CRAFT* ⚠️ *requires fine-tuning CRAFT if it fails to detect characters

Slower Training and Inference* ⚠️ *than DocUFCN